

Mohd Anas

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Summary

2026 B.Tech Computer Science (AI) graduate with experience developing AI/ML solutions across computer vision, predictive modeling, and NLP. Proficient in Python, PyTorch, TensorFlow, Scikit-learn, FastAPI, and model deployment, with experience fine-tuning, evaluating, and deploying machine learning models for production applications.

Skills

Machine Learning: Scikit-learn, Pandas, NumPy, TensorFlow, PyTorch, OpenCV, Ultralytics YOLO

Deep Learning: CNN, Transfer Learning, Object Detection, Computer Vision, Model Evaluation

AI Frameworks: LangChain, LangGraph, RAG, GPT-4, Embeddings, LangSmith, Hugging Face

Databases: PostgreSQL, MongoDB, MySQL, Qdrant, Pinecone, ChromaDB

Tools / Deployment: FastAPI, REST APIs, Docker, Git, Streamlit, Plotly

Programming & Core CS: Python, Java, SQL, OOP, Data Structures & Algorithms

Projects

Fire & Smoke Detection using YOLO11n

Python | PyTorch | Ultralytics YOLO11 | OpenCV | D-Fire Dataset

- Fine-tuned a lightweight YOLO11n model on the D-Fire dataset (21K+ images) for real-time fire and smoke detection, optimizing inference for edge deployment while maintaining high detection performance.
- Evaluated the model using Precision, Recall, mAP, confusion matrix, and qualitative inference on unseen images to validate deployment readiness across diverse fire and smoke scenarios.

AI-Powered Insurance Claims Intelligence Platform

Python | FastAPI | LangGraph | CrewAI | ChromaDB | Hugging Face

- Developed a multi-agent AI pipeline using LangGraph and CrewAI to automate insurance claim processing through document understanding, fraud analysis, policy validation, settlement estimation, and explainable decision workflows.
- Engineered production-oriented AI capabilities including Human-in-the-Loop approvals, semantic memory with ChromaDB, LLM evaluation, configurable guardrails, and FastAPI deployment for transparent and reliable claim processing.

Customer Churn Prediction & Risk Analysis

Python | Scikit-learn | Pandas | NumPy | Gradient Boosting

- Developed an end-to-end customer churn prediction pipeline, performing data preprocessing, feature engineering, exploratory data analysis, and training multiple classification models on 7,000+ customer records.
- Optimized model performance using GridSearchCV and evaluated classifiers using Accuracy, Precision, Recall, F1-Score, and ROC-AUC, achieving **85% accuracy** and **0.82 ROC-AUC**.

Experience

Frontend Web Development Intern

May 2025 – Aug 2025

Gymbrawns Enterprises (Remote)

- Integrated RESTful APIs and optimized asynchronous data flow between frontend and backend services, improving application reliability and data consistency.
- Refactored reusable React components following modular software engineering principles, enhancing maintainability, scalability, and code reusability.

Education

KIET Group of Institutions, Ghaziabad, U.P

Sept 2022 – June 2026

B.Tech in Computer Science & Engineering (AI)

Achievements & Certifications

- Published "CritiqueConnect: Augmenting Human Creativity through AI-Driven Semantic Analysis" at IEEE ICECA 2025.
- DeepLearning.AI: AI Agentic Design Patterns with AutoGen
- AWS Certified Cloud Practitioner (In Progress)